Chapter summary

Measurement

Scientific notation

In scientific notation, very large or very small numbers are written in the form $a \times 10^m$, where $1 \leq a < 10$.

Large numbers will use positive powers of 10;

e.g. 2 350 000 kg = 2.35×10^6 kg

Small numbers will use negative power of 10;

e.g. 0.000000016 = 1.6×10^{-8}

Significant figures

These are counted from left to right, starting at the first non-zero digit.

e.g.

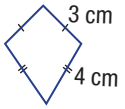
0.00217 has 3 significant figures.

1.07×10^4 has 3 significant figures.

2 403 000 might have 4, 5 or 6 significant figures.

Perimeter

The distance around the outside of a shape.

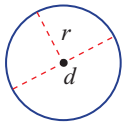


$P = 2 \times 3 + 2 \times 4$
$= 14$ cm

Circumference

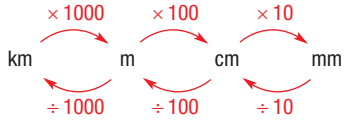
The distance around the outside of a circle.

$C = 2\pi r$ or $C = \pi d$

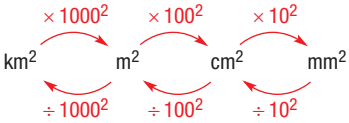


Conversion of units

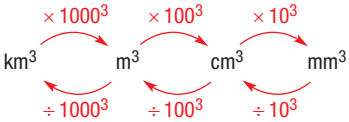
Length



Area



Volume



Area of basic shapes

Square: $A = s^2$

Rectangle: $A = \ell b$

Triangle: $A = \frac{1}{2}bh$

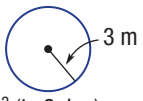
Rhombus: $A = \frac{1}{2}xy$

Parallelogram: $A = bh$

Trapezium: $A = \frac{1}{2}h(a + b)$

Area of a circle

$A = \pi r^2$
$= \pi \times 3^2$

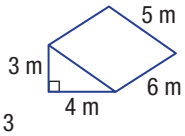


Area = 28.27 m² (to 2 d.p.)

Surface area

Draw a net and sum the surface areas.

Triangular prism



$A = 2 \times \frac{1}{2} \times 4 \times 3$
$+ 6 \times 4 + 6 \times 3 + 6 \times 5$

Surface area = 84 m²

Accuracy

Accuracy of measurement is affected by human error and measuring instruments.

Limits of accuracy give upper and lower boundaries for measurement.

Usually, $\pm$ 0.5 × smallest unit of measurement.

Volume

Prism: $V = Ah$

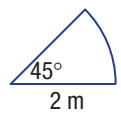
Cylinder: $V = \pi r^2 h$

Capacity: 1 L = 1000 mL

1 cm³ holds 1 mL.

Area of sectors

$A = \frac{45}{360} \times \pi r^2$
$= \frac{1}{8} \times \pi \times 2^2$



Area = 1.57 m² (to 2 d.p.)

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Chapter review

Chapter tests and worksheets can be found in the Teacher Resource Package. Interactive fill-in-the gaps and drag & drop literacy activities can be found in the Interactive Textbook.

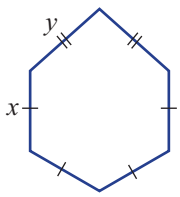
Multiple-choice questions

An online version of this test is available in the Interactive Textbook.

1 The number of metres in a kilometre is:
A 10 **B** 100 **C** 1000 **D** 10 000 **E** 100 000

2 The perimeter of a square with side length 2 cm is:
A 4 cm **B** 8 cm **C** 4 cm^2 **D** 8 cm^2 **E** 16 cm

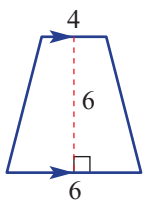
3 The perimeter of the shape shown is given by the formula:
A $x - y$ **B** $2x + y$ **C** $4x + 2y$
D $x - 2y$ **E** $4x + y$



4 A correct expression for the circumference of a circle with diameter 6 cm is:
A $\pi \times 6$ **B** $\pi \times 3$ **C** $2 \times \pi \times 6$ **D** 2×6 **E** $\pi \times 6^2$

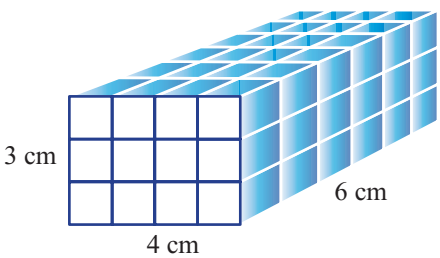
5 The area of a rectangle with side lengths 3 cm and 4 cm is:
A 12 cm^2 **B** 12 cm **C** 7 cm^2 **D** 14 cm **E** 14 cm^2

6 The correct expression for calculating the area of this trapezium is:
A $(6 - 4) \times 6$ **B** $\frac{1}{2} \times 6 \times (6 + 4)$
C $\frac{6}{2} \times (6 \times 4)$ **D** $6 \times 6 - 4$
E $6 \times 6 + 6 \times 4$



7 A sector's centre angle measures 90°. This is equivalent to:
A $\frac{1}{5}$ of a circle **B** $\frac{1}{2}$ of a circle **C** $\frac{3}{4}$ of a circle
D $\frac{2}{3}$ of a circle **E** $\frac{1}{4}$ of a circle

8 The volume of the shape shown is:
A 13 cm^3 **B** 27 cm^3 **C** 72 cm^2
D 72 cm^3 **E** 27 cm^2



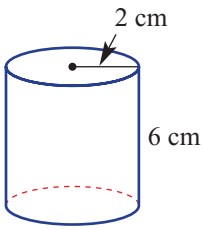
9 The volume of a cube with side length 3 cm is:
A 9 cm^3 **B** 27 cm^3 **C** 54 cm^2 **D** 54 cm^3 **E** 27 cm^2

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Chapter review

10 The capacity of this cylinder is closest to:

A 12 mL **B** 24 mL **C** 75 mL
D 76 mL **E** 750 mL



Short-answer questions

1 Convert these measurements to the units shown in brackets.

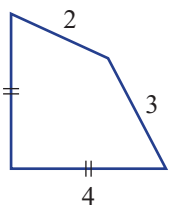
a 5.3 km (m) **b** 27 000 cm^2 (m^2) **c** 0.04 cm^3 (mm^3)
d 1 day (s) **e** 0.125 s (ms) **f** 89 000 000 KB (TB)

2 Give the lower and upper limits for these measurements.

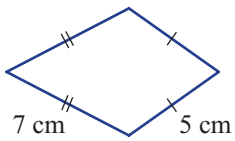
a 6 cm **b** 4.2 kg **c** 15 mL

3 Find the perimeter of these shapes.

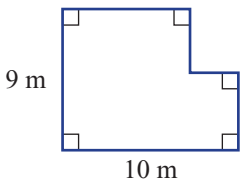
a



b

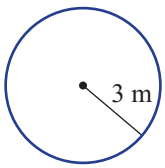


c



4 For the circle, find, to 2 decimal places:

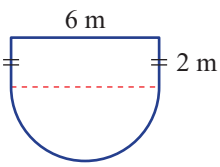
a the circumference
b the area.



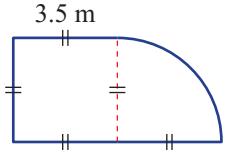
5 For these composite shapes, find, to 2 decimal places:

i the perimeter **ii** the area

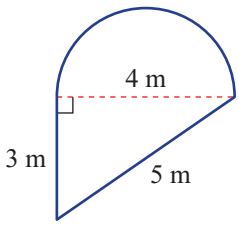
a



b

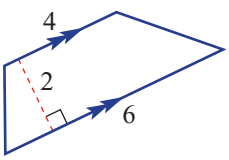


c

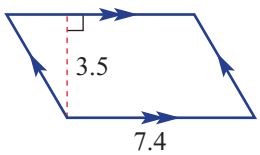


6 Find the area of these shapes.

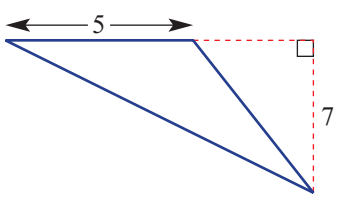
a



b



c

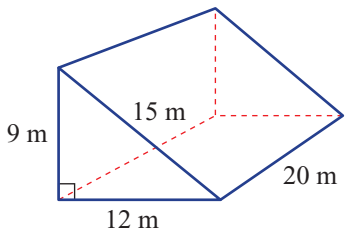


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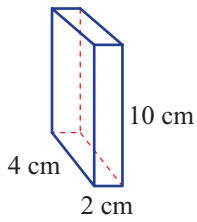
Chapter review

7 Find the surface area of these prisms.

a



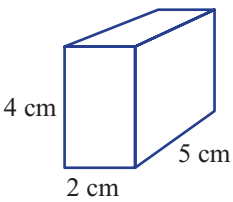
b



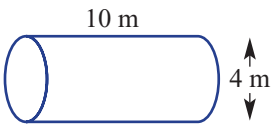
8 **a** Write 232 000 in scientific notation.
b Write 0.000232 in scientific notation.
c Write 4.54×10^6 as a number.
d Write 4.54×10^{-6} as a number.

9 Find the volume of these solids, to 2 decimal places where necessary.

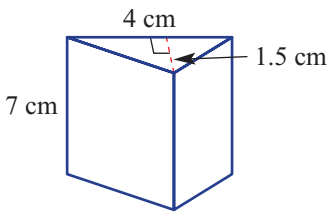
a



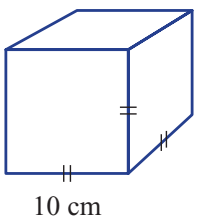
b

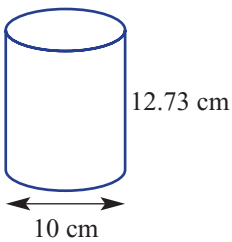


c



10 These two solids are to be filled with water. Which one holds the most?

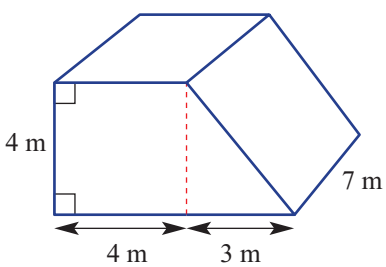




Extended-response questions

1 A cylindrical tank has diameter 8 m and height 2 m.
a Find the volume of the tank, to 2 decimal places.
b Find the capacity of the tank in litres, to 2 decimal places. Note: Every cubic metre holds 1000 litres.

2 A rabbit hutch is to be built in the shape shown.
a Use Pythagoras' theorem to find the slant height of the hutch.
b The hutch will be covered in chicken wire. Determine, in square metres, the amount of chicken wire required. Do not include the base.
c If chicken wire costs $6 per square metre, find the cost of covering the hutch.
d What is the volume of the hutch?



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